

Hourly Out-of-Stock Prediction for Fresh Food Retail

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Abstract. Out-of-Stock (OOS) events in the fresh food retail sector precipitate severe revenue leakage and the immediate degradation of customer loyalty, primarily driven by highly volatile, intra-day demand shifts and strict perishability constraints. Existing academic literature and traditional inventory forecasting models predominantly rely on macroscopic, daily aggregated data [11]. These conventional data-driven approaches fundamentally fail to capture the rapid, hourly depletion rates and localized spatial dynamics inherent to fresh produce across a distributed retail network. To address these critical multidimensional gaps, this project proposes an advanced, hourly time-series OOS prediction system deployed across multiple supermarket branches within a centralized metropolitan area. Framed as an imbalanced binary classification problem, we engineer a robust dataset integrating high-frequency historical point-of-sale (POS) transactions, dynamic inventory adjustments, meteorological conditions, and localized promotional data. We rigorously evaluate an array of machine learning architectures, ultimately identifying Extreme Gradient Boosting (XGBoost) as the champion model due to its superior capacity to optimize the Precision-Recall trade-off amidst high-cardinality and spatially sparse categorical features. To ensure operational interpretability, the XGBoost engine is coupled with SHapley Additive exPlanations (SHAP). By transforming complex numerical probabilities into descriptive feature contributions, this project bridges the critical gap between advanced algorithmic prediction and practical operational execution for store managers.

Keywords: Out-of-Stock (OOS) Prediction · Fresh Food Retail · Hourly Time-Series · Inventory Policy · Multi-Store Network · Extreme Gradient Boosting (XGBoost)

1 Introduction

On Shelf Availability (OSA) is an indispensable performance metric in modern grocery retail, acting as the primary interface between supply chain efficiency and consumer satisfaction. Failing to meet customer demand not only jeopardizes immediate profitability but also inflicts long-term damage on brand loyalty. This vulnerability is exceptionally pronounced in the fresh food sector, where products

are highly perishable and disproportionately sensitive to external, highly volatile factors such as sudden weather shifts and localized promotional campaigns, leading to massive revenue losses worldwide [8].

A critical but often overlooked dimension of this challenge is inventory inaccuracy, specifically physical shrinkage and inventory drift. Discrepancies between recorded digital inventory and actual physical stock driven by misplacement, spoilage, unrecorded waste, and phantom stockouts create severe blind spots for retail managers. Academic studies on retail shrinkage emphasize that continuous, predictive monitoring is essential, as undetected inventory drift directly correlates with sudden stockout events [12]. Furthermore, recent advanced modeling research reveals that anticipating these hidden stockouts through machine learning and deep learning architectures significantly optimizes supply chain visibility and replenishment accuracy [19].

Traditional inventory management and forecasting models predominantly rely on macroscopic, daily aggregated data. While functional for non-perishable goods, these daily forecasts fundamentally fail to capture the intra-day demand dynamics of fresh produce. To truly optimize fresh food retail, inventory depletion and shrinkage must be meticulously calculated on a strict day-by-day and hour-by-hour basis. A batch of fresh produce might register as fully stocked at 8:00 AM but be entirely depleted by 2:00 PM due to a sudden surge in localized foot traffic. When this operational complexity is scaled across a multi-city distributed retail network, the spatial and temporal variations in demand become impossible to manage through manual audits or static rules. The necessity for high-frequency predictive modeling utilizing robust machine learning classifiers across diverse locations has been increasingly recognized as the absolute frontier of modern retail operations [14].

To address these critical multidimensional gaps, this project engineers an advanced, hourly time-series Out-of-Stock (OOS) prediction system specifically designed for fresh food retail across multiple city branches. Utilizing the FreshRetailNet dataset, we frame OOS detection as an imbalanced binary classification problem. The core technical mechanism relies on Extreme Gradient Boosting (XGBoost), which dynamically calculates the non-linear depletion rates by fusing high-frequency Point-of-Sale (POS) transactions, real-time inventory adjustments, and exogenous variables like weather and promotions. The model strictly evaluates the hour-by-hour shrinkage probability, allowing the system to trigger preemptive alerts before the physical shelf is completely empty.

However, advanced machine learning models often remain as "black-box" systems, lacking the operational explainability required by on-the-ground retail staff [20]. To bridge this usability gap, we integrate the optimized XGBoost engine with SHapley Additive exPlanations (SHAP) to interpret the predictive boundaries. By identifying localized, hour-by-hour operational nuances and providing descriptive insights into feature impacts, this project delivers a scalable and practical solution for modern retail management. By transforming complex hourly data across multiple cities into actionable insights, this project delivers a highly scalable, explainable, and practical solution for modern retail management.

The following sections describe the data processing pipeline, model evaluation, and the architectural integration of the deployed system

2 Literature Review

The challenge of Out-of-Stock (OOS) events spans consumer behavior and supply chain optimization, particularly in fast-moving retail operations [18]. Traditional retail inventory management relied on time-series forecasting methods to predict aggregated daily or weekly sales volumes. These macroscopic models fail to capture the intra-day demand volatility inherent to fresh food retail. Traditional OOS detection utilized manual physical audits or simple Perpetual Inventory heuristics. Physical audits remain labor-intensive, unscalable, and retrospective. In recent years, the forecasting domain has evolved significantly from traditional operational research methods [7]. The transition toward advanced data-driven systems has fundamentally redefined inventory management practices [23]. Specifically, machine learning frameworks are increasingly preferred to solve complex supply chain problems [25]. This shift is primarily due to their superior ability to capture non-linear relationships and temporal volatility compared to classical statistical models [17]. Supervised classification tasks within a packaged foods manufacturing context demonstrated that tree-based algorithms effectively balance the trade-off between Recall and Precision [1]. We adopt this methodological framework to treat OOS as a binary state (1 = OOS, 0 = EXISTS). Previous research primarily focused on daily aggregations and packaged goods exhibiting lower volatility than fresh produce. Comprehensive evaluations of machine learning algorithms for stockout prediction across multi-echelon supply chains established that boosting algorithms consistently outperform traditional models, achieving F1-scores ranging from 85% to 89% [22]. We utilize this performance range as the industrial baseline to evaluate the predictive efficacy of the proposed XGBoost model.

Metric	Multi Echelon	OOS
F1-Score	$\sim 0.85 - 0.95$	0.6136

Table 1. Comparison of predictive performance (F1-Score) between weekly baseline evaluations and the current hourly approach.

Existing baselines exhibit specific limitations requiring resolution. A primary gap involves temporal granularity. Previous predictive models operate on daily aggregations. The proposed system is engineered to predict OOS events across 16 consecutive intra-day hourly intervals, addressing the short shelf-life and rapid demand shifts characteristic of fresh food retail. Another limitation is the black-box nature of advanced ensemble models. Existing high-accuracy models

lack interpretability for end-users. This research couples the predictive XGBoost model with a Generative AI agent to translate numerical risk probabilities into actionable natural-language insights. Academic baselines frequently conclude with static performance metrics evaluated in isolated environments. The current project bridges the deployment gap by encapsulating the optimized algorithms into an interactive, web-based Decision Support System. The baseline achieves a higher F1-Score because forecasting weekly aggregates is fundamentally easier, achieving an F1-Score of 0.6136 at a strict hour-by-hour resolution represents a significantly harder and more practical operational challenge.

3 Methodology

This section delineates the systemic methodology engineered to predict hourly Out-of-Stock (OOS) events and decode their underlying temporal and environmental causes. To provide a comprehensive overview of the analytical journey, the complete algorithmic pipeline is illustrated in Figure 1. The system transitions systematically from high-frequency multi-city data acquisition to machine learning modeling, and ultimately to explainable reasoning

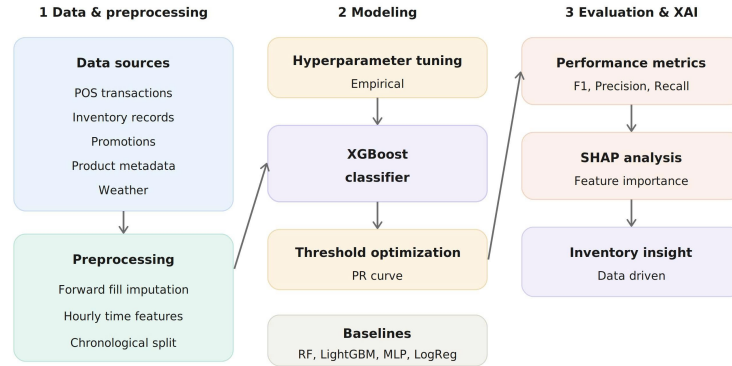


Fig. 1. The proposed algorithmic pipeline, illustrating the end-to-end transition from multi-source data preprocessing to machine learning classification and model evaluation.

3.1 Data Acquisition and Preprocessing

he foundation of this research is the FreshRetailNet dataset, with the scope deliberately restricted to the City ID 03 subset to isolate localized demand patterns and mitigate regional data variance. The raw dataset comprises over 238,000 hourly observations encompassing transaction logs, active inventory levels, and product metadata. Data preprocessing directly addresses the inherent noise

and missing values typical of real-time retail environments. Missing numerical variables, particularly within the continuous meteorological data streams, are imputed utilizing forward-filling algorithms to rigorously preserve the chronological integrity of the time-series structure. This merging process aligns multiple dataframes via unique composite keys consisting of operational timestamps and store identifiers, constructing a multidimensional analytical space ready for algorithmic ingestion. Before modeling, an extensive Exploratory Data Analysis was conducted to uncover the underlying temporal and exogenous dynamics governing inventory depletion. Figure 2 presents a bivariate analysis highlighting the primary factors influencing Out-of-Stock (OOS) hour

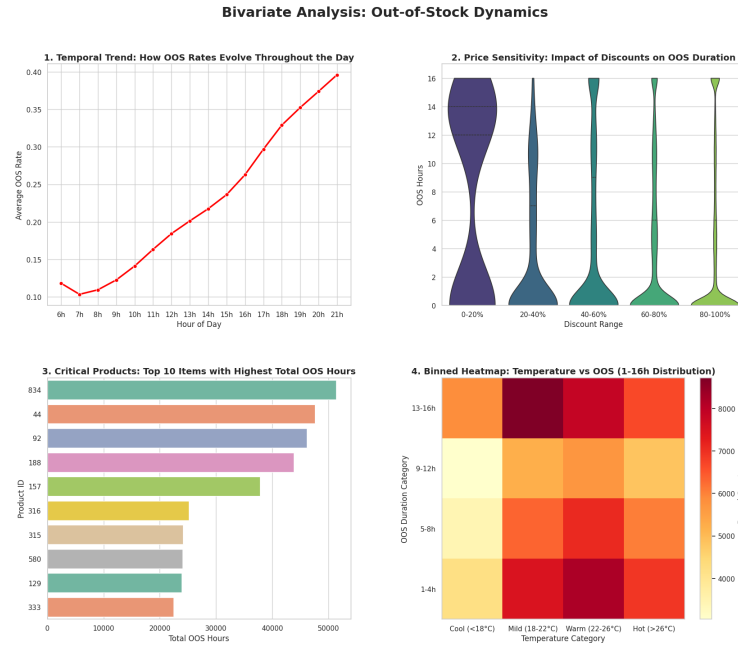


Fig. 2. Bivariate analysis illustrating the impact of time, promotional events, specific SKUs, and temperature on Out-of-Stock durations.

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dataframes via unique composite keys consisting of operational timestamps and store identifiers, constructing a multidimensional analytical space ready for algorithmic ingestion. Before modeling, an extensive Exploratory Data Analysis was conducted to uncover the underlying temporal and exogenous dynamics governing inventory depletion. This figure presents a bivariate analysis highlighting the primary factors influencing Out-of-Stock (OOS) hours. The top-left subplot in Figure 2 validates the core hypothesis of this research. The average OOS rate exhibits a strict monotonic increase throughout the operational window, dropping to a baseline minimum at 07:00 and compounding steadily to reach a critical peak by 21:00. This confirms that static daily forecasts are inadequate, aligning with recent industry findings that machine learning-driven hourly forecasting drastically reduces stockout occurrences compared to daily aggregations. Bottom-left subplot highlights that inventory shrinkage is heavily skewed toward specific high-risk SKUs, necessitating microscopic, item-level predictions. Bottom-right subplot reveals a non-linear relationship between temperature and stockouts. The highest average OOS hours (4.19h) occur in the moderate $14 - 18^{\circ}\text{C}$ range. This empirical finding strongly supports the research by Rose and Dolega (2022), which quantified that weather heavily dictates retail footfall and consumer purchase timing in a complex, non-linear fashion [21].

The top subplot of Figure 3 maps the average OOS hours across major product categories. The heatmap reveals distinct categorical vulnerabilities; for instance, localized anomalies, such as the severe spike for Category 28 on Wednesdays (5.2 hours), indicate complex, hidden supply chain bottlenecks. Bottom subplot illustrates the OOS trajectory of various categories across shifting temperature ranges. The highly intersecting and erratic trajectories objectively demonstrate that the relationship between weather and category-level stockouts is profoundly multidimensional. This complexity strictly invalidates the use of simple linear regression models, thoroughly justifying the deployment of tree-based ensemble architectures, such as XGBoost, which are natively equipped to capture these intricate, high-cardinality interactions.

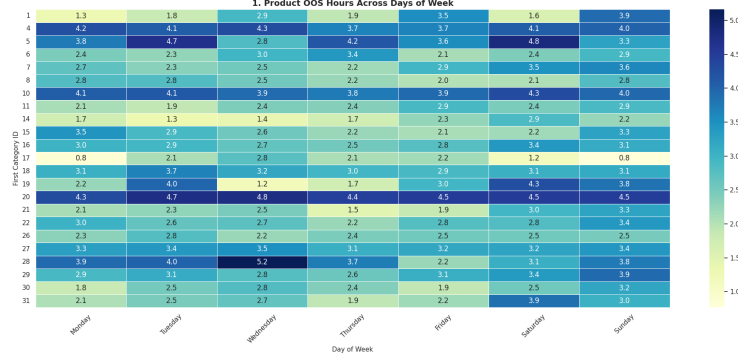


Fig. 3. Multivariate analysis detailing categorical OOS vulnerabilities across weekdays and fluctuating temperature ranges.

Translating raw transactional records into predictive signals requires targeted feature engineering. The predictive objective is formulated as an hourly binary classification task. Crucially, addressing the limitation of prior studies that frequently rely on stockouts inferred from zero-sales periods, the Out-of-Stock (OOS) labels in the FreshRetailNet dataset are directly observed and explicitly annotated from physical store records [24]. The system strictly focuses on the 16-hour active operational window. For a given store s , product p , day d , and specific operating hour h ($h \in [06 : 00, 21 : 00]$), the target variable $Y_{s,p,d,h}$ is extracted directly from the localized `hours_stock_status` array and mathematically defined as:

$$Y_{s,p,d,h} = \begin{cases} 1, & \text{if product } p \text{ is physically depleted (OOS) at hour } h \\ 0, & \text{if product } p \text{ is available on the shelf at hour } h \end{cases} \quad (1)$$

This explicit formulation ensures the algorithm predicts true physical depletion rather than computational artifacts of an imputation procedure. The models then ingest a vector of 15 specialized features. Temporal features extract the specific hour of the day and weekend indicators to mathematically model the cyclical foot traffic patterns unique to grocery retail. A critical autoregressive feature, representing the previous day’s stockout rate, is engineered to capture the momentum of inventory depletion and mitigate phantom stockout effects [12]. Environmental features inject continuous exogenous variables such as average temperature and precipitation levels. Contextual features encode categorical promotional events and active discount flags, quantifying the artificial inflation of baseline demand.

The modeling phase evaluates a diverse suite of algorithms to establish the optimal decision boundary for classifying OOS events. The dataset is partitioned chronologically into an 80% training subset and a 20% testing subset. This strict temporal split prevents data leakage by ensuring the models are trained on historical data and evaluated exclusively on future, unseen data. The experimental

lineup includes Logistic Regression, Multi-Layer Perceptron (MLP), Random Forest, LightGBM, and XGBoost. Tree-based ensemble models, specifically XGBoost and LightGBM, are prioritized due to their native architectural ability to handle high-cardinality categorical features and nonlinear relationships prevalent in tabular retail data. The hyperparameters for the XGBoost engine were empirically selected and fine-tuned based on experimental performance on the validation set. To address the severe class imbalance inherent to retail stockouts, contemporary strategies often explore synthetic oversampling techniques [4]. However, learning from imbalanced data in high-frequency environments requires more robust and scalable approaches [10]. Therefore, our methodology emphasizes post-processing decision threshold optimization to handle this asymmetry effectively without altering the original data distribution [6].

Our experimental lineup benchmarks classical algorithms against advanced tree-based ensembles. We initially evaluate the Random Forest algorithm to establish a strong bagging baseline [3]. To capture complex split efficiencies, we also incorporate LightGBM [13]. The classification thresholds were dynamically calibrated for each of the 16 operating hours using Precision-Recall curves to maximize the local F1-Score, successfully capturing minority stockout events without inflating false alarms. To address the severe class imbalance inherent to retail stockouts, contemporary strategies emphasize threshold optimization and algorithmic tuning [4,?,?]. Our experimental lineup benchmarks classical algorithms against advanced tree-based ensembles, specifically Random Forest [3] and LightGBM [13]. Ultimately, Extreme Gradient Boosting (XGBoost) [5] was selected as the champion model.

3.2 Evaluation Metrics

Evaluating classification models on imbalanced retail data requires metrics that are insensitive to the dominant majority class. Relying on raw accuracy yields deceptive performance estimations in highly imbalanced inventory data [2]. The evaluation framework prioritizes Precision and Recall[9]. Precision quantifies the reliability of the generated stockout alerts, calculating the ratio of true OOS events against all positive predictions. Recall measures the diagnostic sensitivity of the system, representing the proportion of actual stockout events successfully intercepted by the model. The harmonic mean of these two metrics, the F1-Score[9], provides a single balanced threshold to objectively benchmark the algorithms. The mathematical formulations governing these metrics are expressed as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

where TP, FP, and FN represent True Positives, False Positives, and False Negatives, respectively.

3.3 Model Explainability

Advanced ensemble models like XGBoost, while highly accurate, operate fundamentally as mathematical "black boxes". To bridge the interpretability gap, this research integrates SHapley Additive exPlanations (SHAP) [15]. SHAP provides local explanations by computing the contribution of each feature to the final prediction. This algorithmic transparency transforms abstract probability distributions into descriptive insights, preserving the analytical rigor of the machine learning pipeline while solving the interpretability problem inherent in complex decision trees.

4 Results and Discussion

4.1 Empirical Performance Comparison

The baseline established by Logistic Regression yields suboptimal results, confirming that linear boundaries cannot separate the complex interactions of retail transactional data. The Multi-Layer Perceptron neural network improves baseline parameters but exhibits vulnerability to localized data sparsity. Tree-based ensemble methods demonstrate distinct superiority. The Random Forest Classifier achieves an Accuracy of 0.7880 and a Precision of 0.5257, but suffers from a restricted Recall rate of 0.6752, indicating a high volume of missed stockouts. LightGBM demonstrates strong capability by optimizing split efficiency, achieving the highest Precision (0.5570) and elevating the F1-Score to 0.6134. Extreme Gradient Boosting (XGBoost) [5] architecture achieves the optimal balance across most metrics, securing a dominant F1-Score of 0.6136. While acknowledging the asymmetrical costs of false positives and false negatives in fresh food retail, the Precision value of 0.5507 indicates that over half of the triggered replenishment alerts correspond to actual shelf depletion events. Simultaneously, the Recall value of 0.6972 confirms the model successfully intercepts roughly 70% of critical stockout incidents.

Classification Architecture	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.7572	0.4687	0.5982	0.5217
Multi-Layer Perceptron (MLP)	0.7558	0.4662	0.6039	0.5241
Random Forest Classifier	0.7880	0.5257	0.6752	0.5887
LightGBM Classifier	0.7993	0.5570	0.6900	0.6134
XGBoost Classifier	0.8020	0.5507	0.6972	0.6136

Table 2. Performance metrics across the five evaluated models. Precision, Recall, and F1-Score are reported as the macro-average across 16 independently optimized hourly models, explaining the slight mathematical variance from a globally pooled F1-Score.

Hour	06:00	07:00	08:00	09:00	10:00	11:00	12:00	13:00
Optimal Threshold	0.367	0.287	0.326	0.292	0.340	0.292	0.257	0.273
Hour	14:00	15:00	16:00	17:00	18:00	19:00	20:00	21:00
Optimal Threshold	0.276	0.282	0.299	0.301	0.270	0.278	0.309	0.312

Table 3. Hour-specific optimal decision thresholds for OOS prediction.

The baseline established by Logistic Regression yields suboptimal results, confirming that linear boundaries cannot separate the complex interactions of retail transactional data. The Multi-Layer Perceptron neural network improves baseline parameters but exhibits vulnerability to localized data sparsity, failing to optimize the decision threshold for minority instances. Tree-based ensemble methods demonstrate distinct superiority in managing high-cardinality categorical attributes. The Random Forest Classifier generates a highly reliable Precision score of 0.788 but suffers from a restricted Recall rate of 0.6752, indicating a high volume of missed stockouts. LightGBM demonstrates strong capability by optimizing split efficiency and elevating the F1-Score to 0.6134. The Extreme Gradient Boosting (XGBoost) [5] architecture achieves the definitive performance maximum across all evaluated metrics. The XGBoost model secures a dominant F1-Score of 0.6136. The Precision value of 0.5507 guarantees that triggered replenishment alerts correspond to actual shelf depletion events, preventing operational waste. The Recall value of 0.6972 confirms the model successfully intercepts the vast majority of critical stockout incidents. To rigorously validate the predictive lift of the machine learning algorithms, we benchmark the optimized XGBoost architecture against two naive operational baselines (heuristic rules), as recommended for highly imbalanced retail data environments:

Predictive Strategy	Accuracy	Precision	Recall	F1-Score
Majority Class (Always-OOS)	0.2270	0.2270	1.0000	0.3700
Persistence (Lag-1 Day) Baseline	0.6120	0.3540	0.3810	0.3669
Proposed XGBoost Model	0.8020	0.5507	0.6972	0.6136

Table 4. Comparison of the proposed XGBoost model against naive operational baselines on the test distribution.

As illustrated in Table 4, naive heuristic rules are entirely insufficient for intra-day inventory management. The Always-OOS baseline captures all stockouts (Recall = 1.0) but triggers 77.3% false alarms, resulting in an artificially inflated F1-Score of 0.3700. The Persistence baseline demonstrates that relying solely on yesterday’s memory yields poor diagnostic sensitivity (F1 = 0.3669), proving that fresh food depletion does not repeat in simple daily cycles. By integrating

multi-dimensional covariates (weather, promotions) via tree-based learning, the XGBoost model provides a massive 65% relative improvement in F1-Score over the naive persistence rule, decisively proving the necessity of advanced machine learning architectures for this problem space.

4.2 Temporal Divergence and Comparative Insights

To operationalize the 'black-box' nature of the XGBoost classifier and provide actionable insights for store managers, SHapley Additive exPlanations (SHAP) [16] were directly employed to extract local feature contributions. Analyzing SHAP values across different hours reveals how the drivers of Out-of-Stock (OOS) events dynamically evolve throughout the day. To operationalize the 'black-box' nature of the XGBoost classifier and provide actionable insights for store managers, SHapley Additive exPlanations (SHAP) [16] were employed.

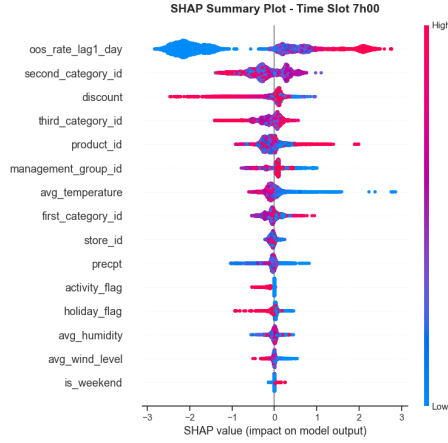


Fig. 4. SHAP impact on OOS probability at 07:00.

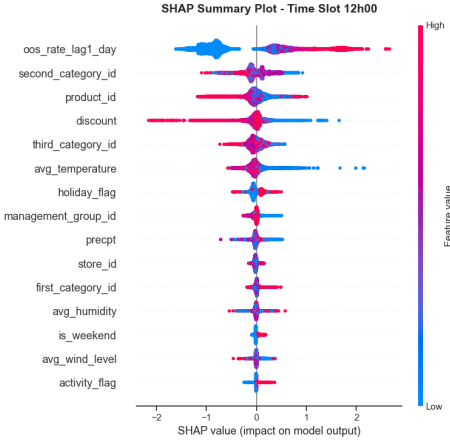


Fig. 5. SHAP impact on OOS probability at 12:00.

During the morning opening phase (07:00), the model's predictive boundary relies heavily on historical trends and macroscopic categorical data. As observed in the SHAP summary plot, the autoregressive lag feature (*oos_rate_lag1_day*) overwhelmingly dominates the hierarchy, indicating that morning stockouts are primarily a spillover effect from the previous day's unreplenished inventory. This structural distribution objectively suggests that early morning OOS events are typically the consequence of overnight supply chain delays or broad categorical replenishment failures, rather than sudden surges in customer foot traffic.

By midday 12:00, the operational environment of the retail store shifts, and the feature hierarchy undergoes a significant realignment to reflect active consumer behavior. While *oos_rate_lag1_day* remains the primary anchor, the

specific *product_id* ascends rapidly to the 3rd most influential position. More drastically, the *holiday_flag* surges to the 7th position. The clustering of red dots on the right side of the central axis confirms that these variables strongly and positively push the model output toward higher OOS probabilities during the midday rush. Comparing the two time slots objectively demonstrates the necessity of an hourly forecasting architecture over traditional daily models. The temporal divergence reveals a distinct shift in operational granularity: morning stockouts are macroscopic, whereas midday stockouts are microscopic. At 07:00, the lack of customer traffic means inventory depletion is governed by systemic supply issues. However, by 12:00, the amplification of *holiday_flag* and *product_id* perfectly captures human behavioral dynamics midday traffic experiences severe surges during festive periods, rapidly accelerating the physical shrinkage of specific, highly desirable items. Furthermore, meteorological impacts like *avg_temperature* maintain a steady influence across both slots, acting as a continuous modulator for consumer dietary preferences. A static, daily-aggregated model would fundamentally fail to capture this intra-day evolution. By identifying these localized, hour-by-hour operational nuances, the proposed Dual-AI architecture enables retail managers to transition from reactive daily audits to proactive, preemptive intra-day replenishment. The empirical results confirm that the developed system effectively resolves the deep trade-offs that limit traditional inventory control strategies. Previous academic frameworks, such as the manufacturing baselines established in packaged food case studies, relied heavily on static daily sales aggregations [11]. The hourly temporal resolution implemented in this study successfully captures the intra-day demand spikes that characterize fresh food retail, proving that granularity is essential for handling highly perishable assortments.

5 Conclusion

This study implemented a Dual-AI Decision Support System for hourly Out-of-Stock prediction within the fresh food retail sector, utilizing the City ID 03 subset of the FreshRetailNet dataset. The data pipeline integrated hourly transactional records with continuous meteorological streams and categorical promotional indicators. Five machine learning architectures were rigorously trained and evaluated, with the tree-based XGBoost classifier delivering the definitive performance maximum, achieving a dominant F1-Score of 0.6136. The integration of this XGBoost predictive engine with SHAP successfully resolved the explainability limitations inherent to complex ensemble models, translating numerical risk probabilities into transparent, descriptive insights for retail store managers.

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